

Rotman

INTRO TO R PROGRAMMING

R Tutorial (RSM358) – Session 2

September 17, 2026 Prepared by Jay Cao / [MDAL](#)

Website: <https://rmdal.github.io/r-intro-2026-fall/>



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Any Questions about Lab 2.3 & A1 Coding?

- Loading the Auto data? (review the 3 options)
 - Option 1: From the ISLR2 library
 - Option 2: From Auto.data (tab separated data)
 - Option 3: From Auto.csv (comma separated data)
- Removing rows with missing data?
 - `na.omit()`

Any Questions about Lab 2.3 & A1 Coding?

- `summary()`, `range()`, `mean()`, `sd()` and dataframe column slicing
 - `Auto['mpg']` vs `Auto$mpg` vs `Auto[['mpg']]`
 - `Auto[c('mpg', 'horsepower')]`
 - `range(Auto['mpg'])` vs `range(Auto$mpg)`
 - `range(Auto['mpg'])` vs `range(Auto[c('mpg', 'horsepower')])` ???
 - How about `mean()`
- Knowing the type of an object is IMPORTANT
 - A function takes in certain types of objects and returns certain types of objects
 - i.e., need to know it to invoke the function correctly
 - Basic Data Structures slides and its companion notebook from last session
 - `str()`, `typeof()`, `class()`

Any Questions about Lab 2.3 & A1 Coding?

- `Auto[1:3]` vs `Auto[1:3, ]`
- `Auto[1:3, ]` vs `Auto[-(1:3), ]` vs `Auto[-c(1:3), ]`
- More on dataframe slicing
 - Basic Data Structures notebooks from last session
- `plot()`
- `pairs()`

Any Questions about Lab 2.3?

- Data Simulation Example 1 (from Lab 2.3)

```
x <- rnorm(50)
y <- x + rnorm(50, mean = 50, sd = .1)
cor(x, y)
```

- Data Simulation Example 2 (from 3.7)

```
set.seed (1)
x1 <- runif (100)
x2 <- 0.5 * x1 + rnorm (100) / 10
y <- 2 + 2 * x1 + 0.3 * x2 + rnorm (100)
```

Lab 3.6 Linear Regression - 1

- Run a regression: `my_lm <- lm(formula = ..., data = ...)`
 - e.g. `lm(formula = y ~ x1, data = my_df)`
- Produce the regression report: `summary(my_lm)`
 - Know how to read test statistics
 - Know how to interpret the estimated coefficient;
- Confidence interval for the coefficients: `confint(my_lm)`

Lab 3.6 Linear Regression - 2

- `plot()`
 - Two variable scatter plot: `plot(x, y)`
 - Regression line: `abline(my_lm)`
 - Post-regression diagnostic plot: `plot(my_lm)`
- `predict(object, new_data, interval, level=0.95)`
 - Confidence interval of $\mathbb{E}(Y)$ at $X = x_0$
`predict(my_lm, data.frame(x1 = (c(5, 10))), interval = "confidence")`
 - Prediction interval of Y at $X = x_0$
`predict(my_lm, data.frame(x1 = (c(5, 10))), interval = "prediction")`

lm() R Regression Formula - 1

my_df

y	x1	x2	x3
18	8	307	130
16	8	304	150
...	...	...	...

lm() Regression Formula	Regression Formula
<code>lm(formula = y ~ x1 + x2 + x3, data = my_df)</code> <code>lm(formula = y ~ ., data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$
<code>lm(formula = y ~ . - x3, data = my_df)</code> <code>lm(formula = y ~ x1 + x2, data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$
<code>lm(formula = y ~ 0 + x1 + x2 + x3, data = my_df)</code>	$Y = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$
...	...

Source: <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/formula.html>

lm() R Regression Formula - 2

my_df

y	x1	x2	x3
18	8	307	130
16	8	304	150
...	...	...	...

lm() Regression Formula	Regression Formula
<code>lm(formula = y ~ x1 * x2, data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon$
<code>lm(formula = y ~ x1 + x2 + x1:x2, data = my_df)</code>	
<code>lm(formula = y ~ x1 + x2 + I(x1 * x2), data = my_df)</code>	
<code>lm(formula = y ~ x1 + I(x1^2), data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \epsilon$
<code>lm(formula = y ~ x1 + log(x2), data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 \ln(X_2) + \epsilon$
...	...

Source: <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/formula.html>